



OPEN Long-wave infrared computational multispectral metasurface and spectral reconstruction method

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We demonstrate a computational multispectral metasurface employing a 3×3 photonic crystal array architecture that operates across the longwave infrared spectrum (8–11.5 μm). The designed structure achieves remarkable optical performance with peak transmittance reaching 75.8% and broadband energy utilization efficiency of 41.37%. Notably, the inter-channel transmittance correlation coefficient of 0.17 indicates superior spectral discrimination compared to conventional grating-based systems. We also considered the angular dependence of the array on the incident light. Additionally, to evaluate the spectral reconstruction performance of the transmission spectra under different photonic crystals, a spectral reconstruction deep learning network was constructed with the mean squared error is 2.86×10^{-3} . This architecture establishes a hardware-algorithm co-design framework for next-generation infrared multispectral systems, demonstrating the potential for integrated superlattice detectors with sub-100 μm^2 pixel pitch, which represents a critical advancement for portable spectroscopic applications.

Multispectral detection technology synergistically combines high spatial resolution with precise spectral discrimination, enabling robust identification and characterization of spectrally ambiguous targets through coordinated analysis of cross-band signatures. This technology effectively distinguishes targets from their backgrounds¹, significantly enhancing detection accuracy and operational effectiveness in military and security environments². Meanwhile, long-wave infrared (LWIR) detection offers strong atmospheric penetration and superior capabilities for identifying camouflaged targets³. Superlattice infrared detectors are gaining considerable attention due to their ability to achieve large-format arrays, high spatial resolution, broad spectral response⁴. Therefore, integrating multispectral metasurfaces onto superlattice infrared detector platforms is a promising choice.

Some reports indicate that the majority of computational multispectral metasurfaces for broad-spectrum detection operate in the visible light range, such as quantum dot arrays⁵, photonic crystal slab arrays^{6,7}, and freeform shaped meta-atoms⁸. In contrast, multispectral filters predominantly employ narrowband filtering metasurfaces^{9–11}, which can operate at the cost of time, space, or energy efficiency. Photonic crystals exhibit periodic potentials to incident electrons, and light modulation is achieved by adjusting the constituent materials and geometry of photonic crystals. Micro-spectrometers based on photonic crystals have been integrated on-chip in the visible light range¹², but there are few studies in the LWIR range.

Based on computational spectral demodulation at specific bands, real-time multispectral characteristics can be obtained. Studies have shown that combination of random transmission spectra with spectral information processing algorithms such as least squares^{13,14}, CS theory^{15,16}, or deep learning^{17–19} can provide highly accurate spectral recognition capabilities. Compared with spectral reconstruction algorithms based on least squares minimization, the main advantage of deep learning methods is the ability to handle large amounts of complex nonlinear data.

In this study, we present a computational-type multispectral metasurface utilizing photonic crystals specifically designed for long-wavelength infrared applications, coupled with a novel deep learning framework for spectral reconstruction. Through systematic simulation analyses of photonic crystal parameters, we achieved an optimized 3×3 array configuration demonstrating enhanced average energy utilization efficiency compared to conventional designs. The engineered photonic architecture was developed through iterative parameter optimization, incorporating both structural and material considerations. Furthermore, we propose an advanced deep learning network employing a neural architecture with spectral-spatial feature fusion, capable of reconstructing spectral information with high fidelity (achieving $<20\text{ nm}$ resolution in the 8–11.5 μm range).

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This integrated photonic-computational platform demonstrates remarkable spectral reconstruction accuracy while maintaining compact form factor advantages essential for infrared sensing applications.

Methods

Structure and incident-angle-dependent response

Propose a schematic design for a 3×3 photonic crystal array (as illustrated in Fig. 1), with material selection prioritizing high optical transmittance through the use of gallium antimonide (GaSb) as the substrate medium. The bottom of the metasurface is intended to incorporate a Type-II InAs/GaSb superlattice LWIR photodetector, which operates at 77 K. This structure offers flexible bandgap engineering and lays a solid foundation for the development of high-precision, energy-efficient computational multispectral detectors in the LWIR range. By adjusting the lattice constant and pore size of photonic crystals, bandgaps may be produced within a specific wavelength range²⁰. In order to achieve wideband transmission, this research aims to circumvent these bandgap regions within a specific wavelength range by mapping the position and size of photonic bandgaps in various photonic crystal slabs. The photonic bandgap position map of the crystal is obtained by scanning either the ratio of pore radius to lattice constant or the ratio of thickness to lattice constant. We will simulate the band structure of air columns in a dielectric substrate, selecting the non-bandgap region where the dielectric constant ratio is 13.7:1. The simulated structure, as illustrated in Fig. 1, the intermediate layer comprises a photonic crystal slab featuring a 3×3 array of photonic crystals with systematically varied dimensions. The incident light is decomposed into its constituent electric field and magnetic field components. The first photonic crystal array configuration is geometrically defined with the following parameters: lattice constant (a) spanning $3.6\text{--}12\text{ }\mu\text{m}$, cylindrical cavity radius (r) ranging $0.7\text{--}2.1\text{ }\mu\text{m}$, and a uniformly maintained slab thickness (h) of $1.89\text{ }\mu\text{m}$. The electromagnetic characteristics of the three-dimensional architecture are numerically investigated using Rigorous Coupled Wave Analysis (RCWA)^{21,22}, a full-wave electromagnetic analysis technique. This computational approach enables the acquisition of spectral response characteristics modulated by geometrical variations including hole dimensions (radius), lattice spacing parameters, and metallic layer thicknesses, while simultaneously calculating the transmission spectra through systematic electromagnetic field solutions across the operational wavelength range. Incorporate the obtained transmission spectra into the Pearson correlation coefficient calculation to determine the structural parameters with lower correlation coefficients. The photonic crystal array with lattice constant and pore radius of $a = 4.4\text{ }\mu\text{m}$, $r = 0.89\text{ }\mu\text{m}$ and its transmission spectrum are shown on the right side of Fig. 1.

The 3×3 photonic crystal array demonstrates nine distinct transmission spectra across the $8\text{--}11.5\text{ }\mu\text{m}$ wavelength range, as illustrated in Fig. 2. To evaluate the energy utilization efficiency of these photonic crystal configurations, the spectral energy distribution within designated wavelength bands can be quantitatively analyzed by calculating the integrated transmission energy derived from spectral measurements. This analytical approach enables systematic comparison of energy harvesting performance across different array geometries through spectral data processing. This can be achieved by integrating the radiation intensity as a function of wavelength, as shown in Eq. (2).

$$E_{eff} = \int_{8 \times 10^{-6}}^{11.5 \times 10^{-6}} I(\lambda) d\lambda. \quad (1)$$

Where E_{eff} is the transmitted energy in the wavelength range 8 to $11.5\text{ }\mu\text{m}$, and $I(\lambda)$ is the radiation intensity at wavelength λ . As shown in Table 1 and Table 2, the results show that the area enclosed by the transmission spectra and the wavelength axis, representing the energy utilization rate (EUR), the area shaded beneath the curve corresponds to the EUR in Fig. 2, ranges from 21.4% to 47.2%, with an average of 32.7%. The designed

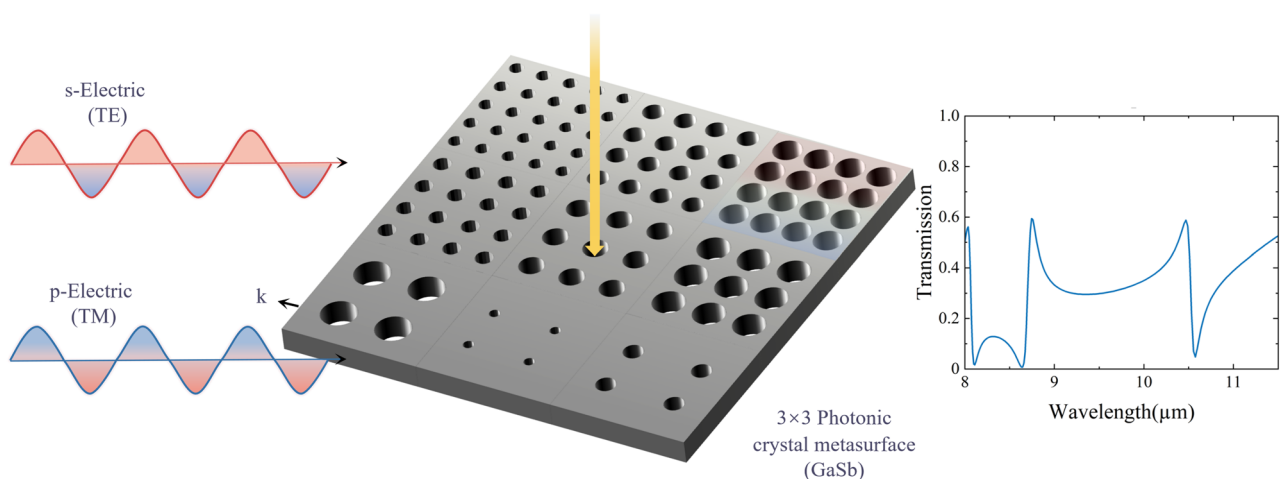


Fig. 1. Schematic diagram of a multispectral metasurface composed of a 3×3 photonic crystal array, demonstrating incident light modulation and corresponding transmission spectral characteristics.

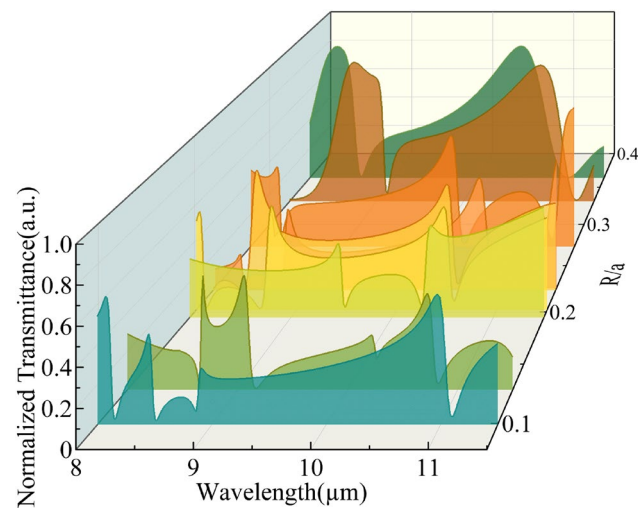


Fig. 2. Transmission spectra of 9 different photonic crystal groups.

Seq.	a (μm)	r (μm)	h (μm)	θ (°)	EUR (%)
a	5.75	1.9	1.89	0	47.22
b	5.75	2.09	1.89	0	45.60
c	4.4	0.89	1.89	0	30.30
d	3.6	0.7	1.89	0	32.57
e	4.4	1.2	1.89	0	33.04
f	4.4	0.98	1.89	0	32.06
g	10	0.99	1.89	30	28.16
h	12	1.53	1.89	45	23.59
i	5.806	1.307	1.89	60	21.35

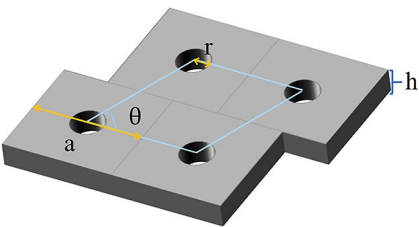


Table 1. Parameters and energy utilization rates of the first group of photonic crystal slab.

Seq.	a (μm)	r (μm)	h (μm)	t (μm)	θ (°)	EUR (%)
a	20	6	29	1	0	47.16
b	20	9	29	1	0	65.24
c	20	8	29	1	0	60.22
d	20	9.72	29	1	0	72.44
e	20	9.3	29	1	0	68.75
f	10	3.1	29	1	0	32.17
g	10	4.55	29	1	0	25.72
h	12	4.87	29	1	0	57.13
i	10	2	29	1	45	15.53

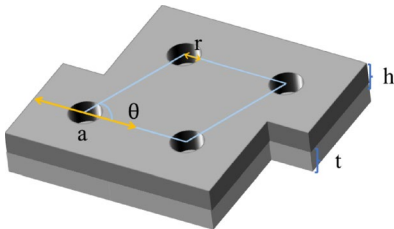


Table 2. Parameters and energy utilization rates of the second group of photonic crystal slab.

narrowband filter exhibits a full width at half maximum (FWHM) of <5% relative to its central wavelength ($\Delta \lambda/\lambda_0 < 0.05$), while representing a $2.4\times$ enhancement over conventional dielectric filters (typically 5–15% efficiency)^{23,24}. The influence of the presence or absence of a substrate was also considered. Two distinct photonic crystal arrays were obtained: the first without a substrate, and the second with a 1 μ m-thick substrate. To calculate the Pearson correlation coefficient²⁵ using the NumPy library in Python, thereby selecting the photonic crystal structure with the minimum correlation coefficient between transmission spectra. The absolute value scatter of the correlation coefficients for photonic crystal arrays is shown in the right panel of Fig. 3. The left panel of Fig. 3 presents a heatmap of the correlation coefficients, with the axes representing different photonic crystal arrays. The mean absolute value of the correlation coefficients between photonic crystal arrays is 0.1697, with the maximum value being 0.3315.

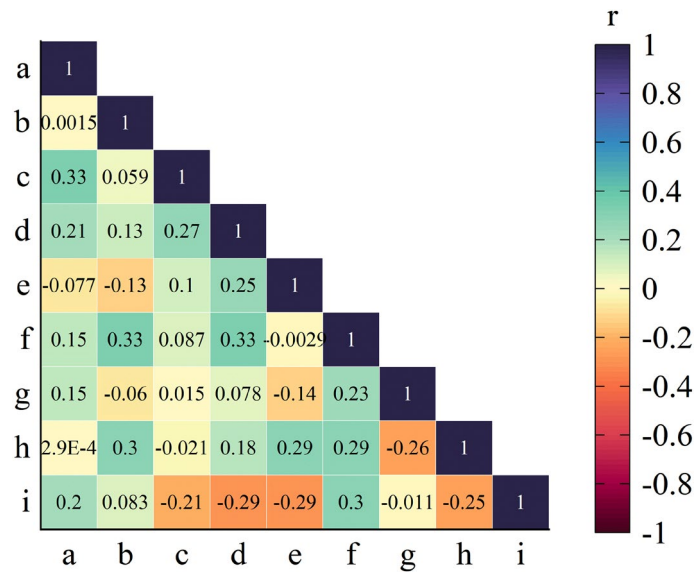


Fig. 3. Pearson correlation coefficients between different photonic crystals.

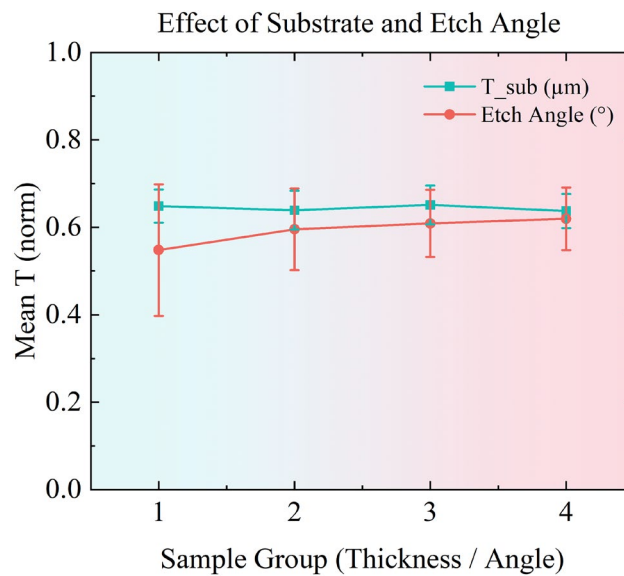


Fig. 4. Effect of substrate thickness and etching angle on transmission spectrum response.

Meanwhile, to evaluate the influence of substrate thickness and the etch angle (defined as the angle between the sidewall and the horizontal plane) on the transmission spectra, arrays with identical lattice constants and hole diameters were compared. As shown in Fig. 4, we examined the transmission responses for substrate thicknesses of 2 μm , 5 μm , 10 μm , and 15 μm , and undercut angles of 26.8°, 34.9°, 45.53°, and 61.23°, respectively. The corresponding mean values and standard deviations were calculated. The minimum observed errors were 0.03898 and 0.07166 for thickness and undercut angle, respectively, indicating that these variations have only a minor effect on the final spectral response.

Further investigation into the physical mechanism of multi-spectral metasurface modulation revealed distinct electric field distributions within the 8–11.5 μm wavelength range. Under TM-polarized normal incidence, the transmission responses of both TE (s-polarized) and TM (p-polarized) waves were analyzed across incident angles θ ranging from 0° to 60°, as shown in Fig. 5. A wide angular range of incident-angle measurements for both TE and TM polarizations was demonstrated. Compared with the first photonic crystal array configuration, the second array incorporated practical design considerations through optimized structural parameters: an increased cavity radius-to-lattice constant ratio (r/a) and enhanced thickness (h) while maintaining fixed height dimensions. Notably, a 1 μm GaSb substrate was integrated into the modified design. The electric field distribution indicates that the substrate and air form a Fabry-Perot resonant cavity, and a distinct resonance

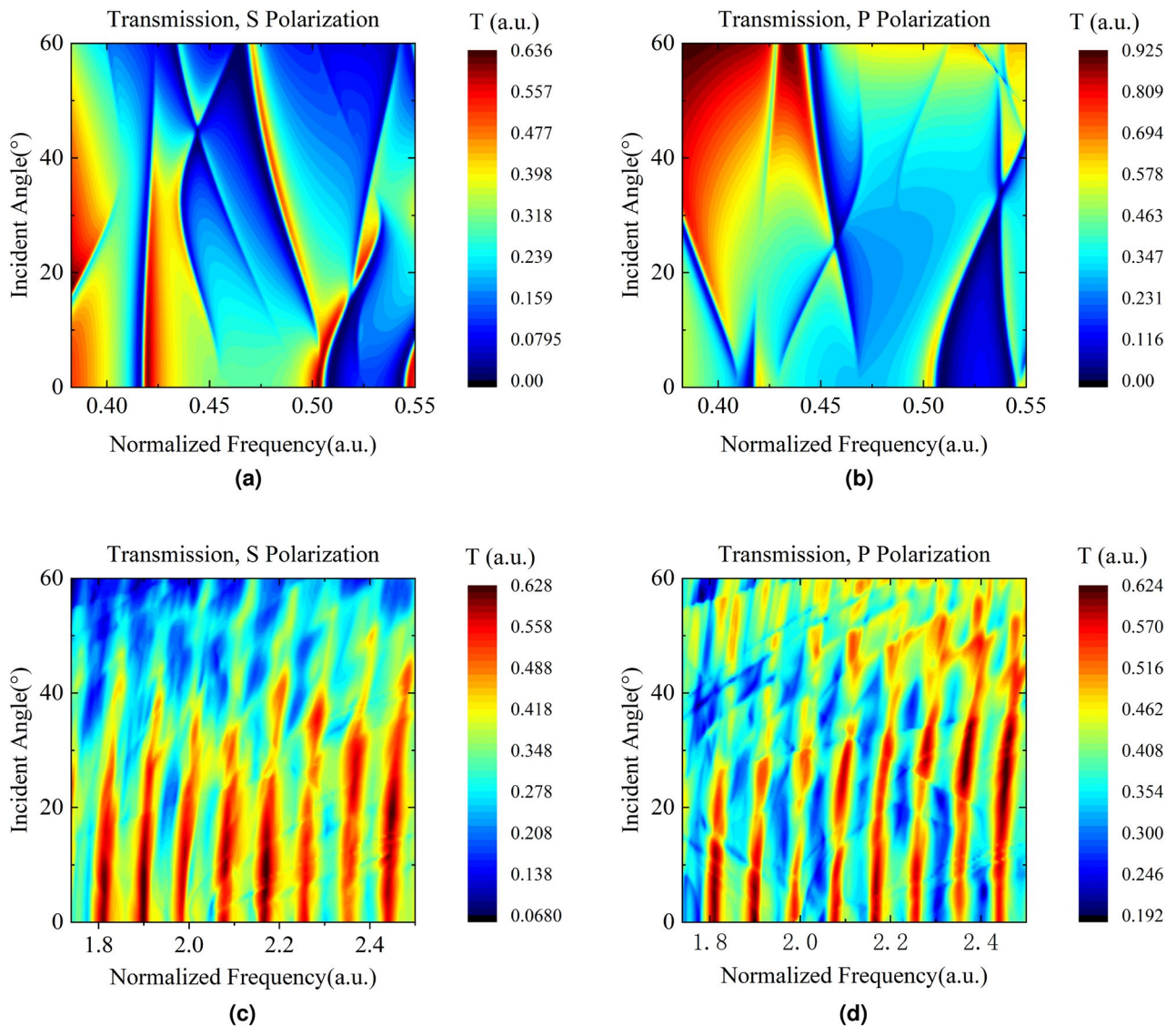


Fig. 5. Transmission of S-wave polarization Spectral responses of two different photonic crystal arrays for S-wave and P-wave in the wavelength range of 8–11.5 μm , with incident angles ranging from 0° to 60° .

peak can also be observed in the second transmission spectrum. As shown in Fig. 5, it can be observed that the sensitivity to the incident angle in the second type of photonic crystal array is lower than that in the first type, indicating that it is largely unaffected by the incident angle.

Isofrequency contours^{31–33} serve as a valuable analytical tool for characterizing photonic crystal behavior by systematically mapping all permissible wave vectors and their associated frequencies. A perfectly circular isofrequency contour indicates ideal isotropic light propagation, where directional dependence is eliminated. The comparative plotting of multiple contours, as illustrated in Fig. 6, enables direct visualization of frequency-dependent contour evolution, providing critical insights into photonic structure modifications. As a diagnostic tool, isofrequency diagrams effectively evaluate light propagation isotropy in photonic crystals. From the Brillouin zone, we can determine the relative amplitudes or intensities of spatial harmonics; these modes are not isolated and are composed of Bloch waves within the photonic crystal. In a uniform medium, the reciprocal lattice constant is zero, meaning no Bloch modes^{34,35} are found. Their circular symmetry—independent of polarization states and propagation directions—serves as a key metric for optical isotropy. Comparative analysis in Fig. 6 demonstrates enhanced isotropy in the right panel, where contours approach near-ideal circular geometry compared to the distorted profiles observed in the left panel.

Spectral reconstruction

To simulate realistic measurement conditions, we introduced various types of noise into the ideal incident spectrum, including but not limited to additive white Gaussian noise (AWGN), thermal noise, shot noise, and instrumental drift. Specifically, AWGN was set with a signal-to-noise ratio (SNR) of 30 dB, thermal noise was simulated at a temperature of 77 K, the photon count rate was set to 10^3 , and the drift amplitude was set to

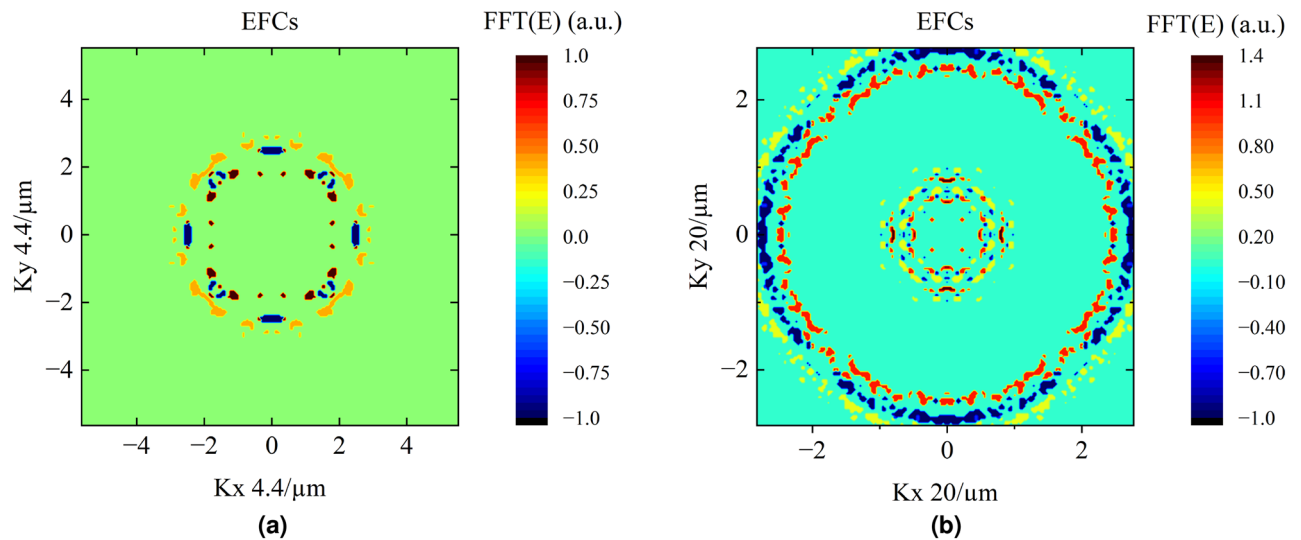


Fig. 6. Isofrequency contours for two sets of 3×3 photonic crystal arrays. The left image represents the first set of photonic crystal arrays, while the right image represents the second set.

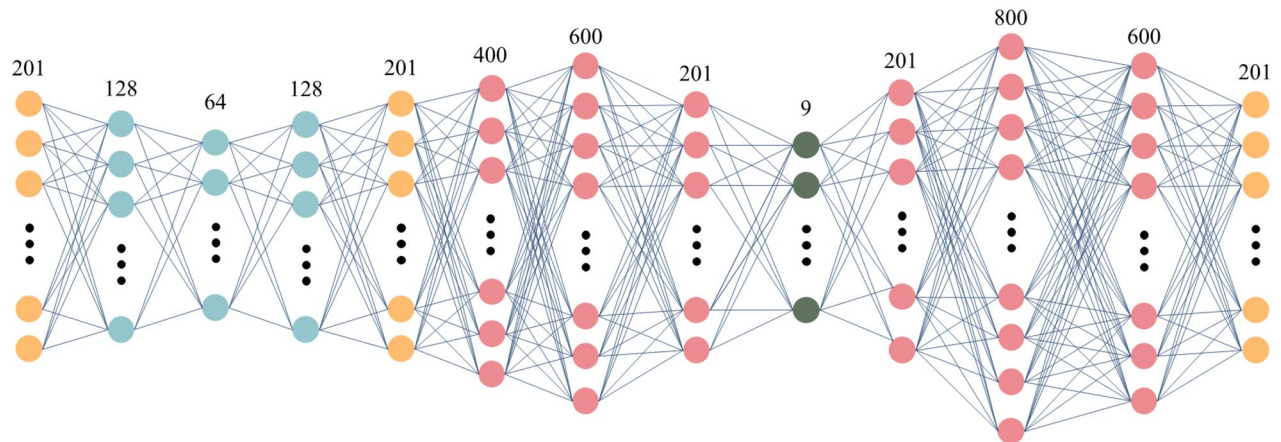


Fig. 7. Transmission spectrum reconstruction prediction algorithm.

0.05. We construct an incident spectrum spanning the wavelength range of $8\text{--}11.5\ \mu\text{m}$ by synthesizing multiple randomly generated Gaussian functions. To investigate the impact of various types of noise on the network architecture, each noise component was independently introduced into the model. The influence of different noise sources on the reconstruction results was evaluated, enabling an analysis of the model's robustness under diverse noise conditions. The spectral reconstruction is subsequently performed through a machine learning approach utilizing a Back Propagation (BP) neural network. This architecture includes a training and testing dataset of $N = 5000$ incident spectra. As shown in the blue region of Fig. 7, the designated DenoisingModule serves as a spectral denoising component based on an autoencoder architecture. This module incorporates several key elements: the Rectified Linear Units (ReLU) introduce nonlinearity into the network and helps prevent gradient vanishing, thereby enhancing the model's ability to learn complex mappings. Dropout layers are applied to randomly deactivate 30% of the neurons during training, serving as a regularization technique to prevent overfitting and improve the network's generalization. Finally, a Sigmoid activation function is used to reconstruct the spectral data in a normalized form. In Fig. 7, the BP-net architecture is as follows: 201-LR-LL-400-LR-LL-600-LL-201-LL-9-LL-201-LR-LL-800-LR-LL-600-LR-LL-201, where LL represents a linear layer, LR denotes the leaky rectified linear unit, several dropout layers are also included. The formula for spectrum reconstruction is as follows:

$$E_i = \int_{\lambda_1}^{\lambda_2} T_i(\lambda) I_n(\lambda) + n_i d\lambda \approx \sum_{j=1}^N T_i(\lambda_j) I_n(\lambda_j) + n_i d\lambda. \quad (2)$$

Where $In(\lambda)$ is the incident spectrum, $T_i(\lambda)$ is the transmission matrix, n_i represents experimental noise, and λ_2 to λ_1 ranges from 8 to 11.5 μ m, N is the number of points (201), and k is the number of different transmission matrices (9).

Therefore, the simulation of the incident spectrum modulation process is described by $E = TIn + n$, where In is the incident spectrum, T is the transmission matrix, E is the detected signal, and n is the simulating noise matrix. The loss function is:

$$loss =_{In}^{min} \|E - TIn\|_2^2 + k \|n_i In\|_2^2. \tag{3}$$

Moreover, some studies have shown that iterative reconstruction algorithms often incorporate constraint terms into the loss function during optimization. However, selecting a single weighting parameter for the constraint typically results in a trade-off between preserving spectral details and reconstructing broadband spectra. In contrast, deep learning-based spectral reconstruction methods can effectively overcome this limitation. As a result, they are capable of achieving ultra-high spectral resolution, as demonstrated in Table 3. Comparative analyses show that Tikhonov regularization (TR), CS, recursive least squares (RLS), and DL methods all perform well under oversampling conditions^{26,30}. However, in undersampling scenarios, DL methods exhibit superior performance, significantly outperforming CS in reconstructing dense spectral features^{6,27,29}. The transmission efficiency of the multilayer film¹⁸ and the Fabry–Perot cavity²⁸ is relatively low. Therefore, this study effectively balances the requirements of broadband spectral coverage and reconstruction accuracy, achieving favorable results in both aspects.

The entire neural network is composed of multiple linear layers, where the incident spectrum is multiplied or mapped with the transmission matrix. The noise is simulated using additive white Gaussian noise, given by n , where σ represents the signal-to-noise ratio, and follows a standard normal distribution. The training layer consists of three linear layers. The input dataset for the neural network includes 5000 incident spectra, simulated using discretized Gaussian functions. These incident spectra are generated by adding a series of Gaussian distribution functions with different peak heights, center coordinates, and half-widths. Each generated spectrum is then normalized by its height. The peak positions are randomly selected from the range of 8 to 11.5 μ m, with intervals of 17.5 nm. The linewidths are randomly distributed within the range of 0.175 μ m to 0.7 μ m. Similarly, reconstructing the spectra at different noise levels reveals that, under high noise conditions, the 3×3 array of different photonic crystal patterns performs better overall compared to the 2×2 array of different photonic crystal patterns. For the purpose of wide-band infrared detection, it is essential to have transmission spectra with low correlation at the spectral wavelength range of 8–11.5 μ m to obtain rich spectral characteristics.

After training and testing cycles, the loss values for the training and testing datasets were reduced to 2.86×10^{-3} and 3.93×10^{-3} , respectively. The statistical distribution of mean absolute error (MAE) and loss function value of the reconstructed spectra and actual spectra of different photonic crystal arrays were calculated. The mean square error (MSE) between the reconstructed spectrum and the actual spectrum is defined as follows:

$$MSE = \frac{1}{k} \sum_{i=1}^k (I_i - \tilde{I}_i)^2. \tag{4}$$

Where k represents the number of discrete data points in the transmission spectrum, which is 201. I_i denotes the true values of the incident spectrum, and $\tilde{I}_i$ represents the spectrum values reconstructed by the network model.

Synthetic spectral signals corrupted by various types of noise were generated to evaluate the performance of the proposed denoising framework. The architecture comprises a denoising autoencoder as a preprocessing module, followed by a primary neural network (Bp-net) for spectral reconstruction. To promote the smoothness of the reconstructed spectra, a Total Variation (TV) loss term was incorporated into the objective function. Experimental results demonstrate that the model achieves robust denoising performance, with an average MAE of 0.01936 and MSE of 0.0958 across different noise scenarios in Fig. 8. These low error metrics indicate that the model effectively mitigates the influence of noise, preserving the structural integrity of the spectral signals.

The low loss values indicate that the training process of Bp performed well in Fig. 9a. After training, the specific error distribution of the test dataset was analyzed. Figure 9b, c present a comparison of the results,

Spectral-selective structure	Reconstruction method	Reconstruction accuracy	Transmission efficiency	Detector type	Spectral range (μ m)	Spectral resolution (nm)	Spatial resolution (μ m)	Ref.
Plasmonic metasurface	RLS	MAE = 0.066 to 0.094	–	IR microscope	1.5–1.9	–	–	26
Plasmonic nanohole arrays	CS	MSE = 6.1×10^{-3}	60%	CMOS	0.5–0.7	0.19	–	27
Fabry–Perot cavity	–	–	35%	Bolometric IR detector	16	–	–	28
Photonic crystal slab	CS	MSE < 0.05	33%	CMOS	0.55–0.75	1	200	6
Photonic crystal slab	CS	–	–	CMOS	0.4–0.7	1	–	29
This work	DNN	MSE = 2.86×10^{-3} to 3.93×10^{-3}	41.37%	InAs/GaSb type-II superlattice	8–11.5	17.5	20	–
Multilayer film	DNN	MSE = 3.2×10^{-3} to 5×10^{-6}	21%	CCD	0.4–0.7	5.2	–	18
Black phosphorus	TR	–	47%	Black phosphorus	2–9	420	–	30

Table 3. Comparative performance analysis of recent computational spectral reconstruction methods.

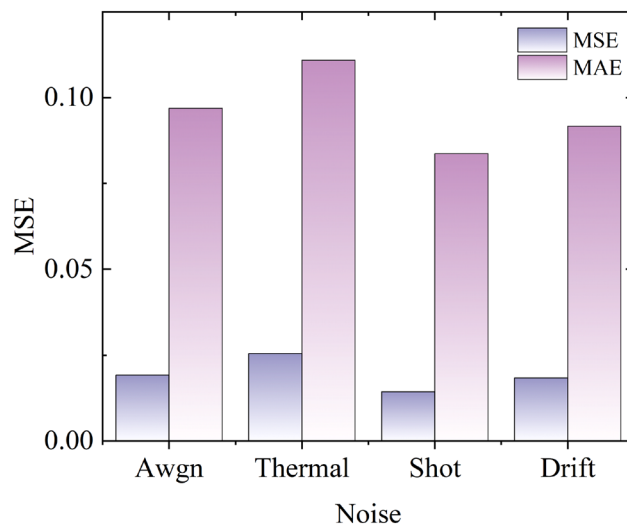


Fig. 8. Effect of different types of noise on deep learning reconstruction of photonic crystal arrays: MSE and MAE between ground truth and reconstructed spectra.

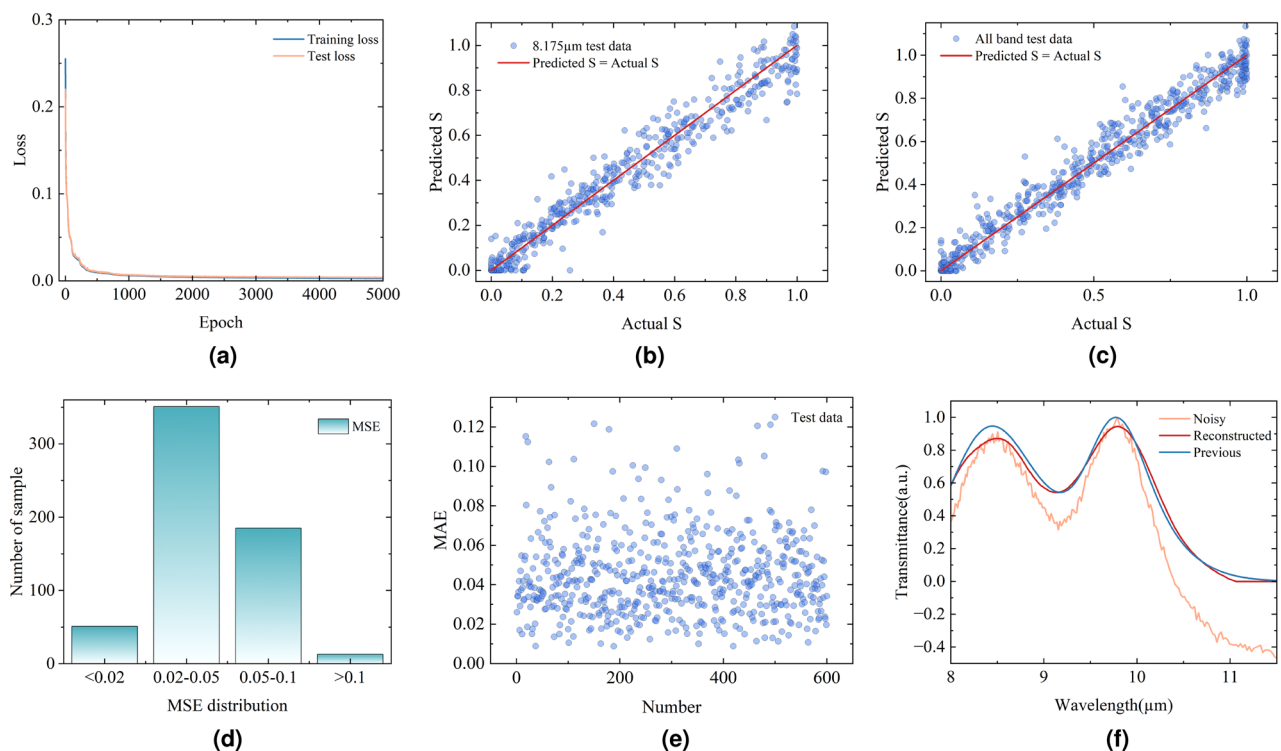


Fig. 9. Deep learning results for photonic crystal arrays: (a) the loss function curve, (b) the scatter plot of the reconstructed spectrum and the true spectrum at a wavelength of 8.175 μm , (c) the scatter plot over the entire wavelength range, respectively, (d) the statistical distribution of test set samples within different MAE ranges, (e) the MAE between the true spectrum and the reconstructed spectrum, and (f) the noisy spectrum, previous spectrum and reconstructed spectrum.

illustrating the distribution of the relationship between the predicted reconstructed spectra and the ground truth. Using MAE as the evaluation metric, scatter plots of the predicted versus true values are shown. The error distribution is visualized for both single-band and full-spectrum cases. As depicted in the figures, the scatter points are closely clustered around the ground truth, with a MSE of 0.0207, indicating that the model achieves high prediction accuracy. As shown in Fig. 9d, it can be seen that most sample data have an MAE error below 0.05, with only 2.17% of the dataset samples having an MAE exceeding 0.1. The MSE of the test dataset

is 2.86×10^{-3} , corresponding to a mean absolute error ($\text{MAE} = \frac{1}{k} \sum_{i=1}^k (I_i - \tilde{I}_i)^2$) of 0.0442 in Fig. 9e. If an MAE below 0.05 indicates high prediction accuracy, then Bp achieves 67% accuracy. When the MAE threshold is relaxed to below 0.1, the prediction accuracy of Bp increases to 97.83%. These results demonstrate that the MAE distribution is well-formed and reliable.

To more rigorously demonstrate the predictive capability of the Bp-net model, a representative sample from the test dataset was randomly selected for detailed comparison, as illustrated in Fig. 9f. The figure presents the true spectrum, the noisy input spectrum, and the reconstructed spectrum generated by the model. Notably, the reconstructed spectrum exhibits a high degree of fidelity to the true spectrum, with significant overlap and a comparatively low MSE. These results substantiate the robustness and accuracy of the proposed neural network approach.

Conclusion

We present an innovative multispectral metasurface architecture integrated with a computational reconstruction framework, specifically engineered for long-wave infrared spectral manipulation across the 8–11.5 μm operational band. Suitable structural parameters were screened based on the correlation coefficient, and the transmission matrix was fed into the reconstruction neural network. The reconstruction errors after training in the region of $[2.86 \times 10^{-3} \sim 3.93 \times 10^{-3}]$, with average energy utilization rates of approximately 32.7% to 49.37%. Deep neural networks, as a data prediction method, significantly improves design efficiency and computational resources. Our results also demonstrate that this design can achieve the purpose of optimizing random multispectral metasurfaces.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

The research was designed by Shang Wang. Sampling, experiment and data collection were done by Shang Wang. Experiment and data collection were done by Shang Wang, Lidan Lu and Lianqing Zhu performed the statistical analysis. Shang Wang wrote the manuscript; the final manuscript was investigated and approved by all the authors.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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